

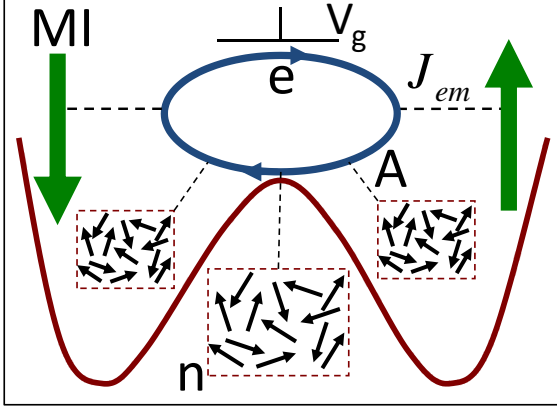


FIG. 1: Schematics of two spin qubit unit in a DQD considered here. Each spin qubit is represented by a MI. Interaction between qubits is mediated by the two electrons; the electron MI coupling coefficient is denoted by J_{em} . As interaction of MIs with nuclear spin bath is negligible, the dominant noise source is electron mediated coupling between MIs and nuclear spin bath; the electron nuclear spin bath coupling coefficient is given by A . The interaction strength between the MIs and between MI and nuclear spin bath depends on the singlet-triplet electron energy gap Δ_e , which can be controlled by external electric gate voltage (V_g), magnetic field (B), and shape/size of the DQD. This allows active noise engineering and optimization of the two qubit system.

between the nuclear spins of MI and host nuclei has no influence on our system.

Electrons confined in quasi-two-dimensional quantum dots in a uniform perpendicular magnetic field can be described by the effective mass Hamiltonian

$$H_e = \sum_{i=1}^N (T_i + Z_i) + \frac{e^2}{2\epsilon} \sum_{i \neq j} \frac{1}{|\vec{r}_i - \vec{r}_j|}, \quad (1)$$

where $T = 1/(2m^*) \left((\hbar/i)\vec{\nabla} + (e/c)\vec{A}(\vec{r}) \right)^2 + V(\vec{r})$ is the single electron Hamiltonian in an external magnetic field $\vec{B} = B\hat{z}$, perpendicular to the plane of the confining potential. Here $(\vec{r}) = (x, y, z)$ describes the electron position, $V(\vec{r})$ denotes the quantum dots confining potential, and $\vec{A}(\vec{r}) = (1/2)\vec{B} \times \vec{r}$ is the vector potential. Further, m^* is the conduction-electron effective mass, $-e$ is the electron charge, and ϵ is the host semiconductor dielectric constant. Finally, $Z_i = (1/2)g_e\mu_b S_{zi}B$ determines the Zeeman spin splitting, g_e is the electron g-factor in host semiconductor, μ_b refers to the Bohr magneton, and S_{zi} represents the z -Pauli matrix of electron i .

The single particle eigenvalues ($\epsilon_{\alpha\sigma}$) and eigenvectors

($\varphi_{\alpha\sigma}$) are calculated by discretizing $T + Z$ in real space, and diagonalizing the resulting matrix. By using the creation (annihilation) operators $c_{\alpha\sigma}^\dagger$ ($c_{\alpha\sigma}$) for an electron in a non-interacting single-particle (SP) state $|\alpha, \sigma\rangle$, the Hamiltonian of an interacting system in second quantization can be written as

$$H_e = \sum_{\alpha} \sum_{\sigma} \epsilon_{\alpha\sigma} c_{\alpha\sigma}^\dagger c_{\alpha\sigma} + \frac{1}{2} \sum_{\alpha\beta\mu\nu} \sum_{\sigma\sigma'} V_{\alpha\sigma, \beta\sigma', \mu\sigma', \nu\sigma} c_{\alpha\sigma}^\dagger c_{\beta\sigma'}^\dagger c_{\mu\sigma'} c_{\nu\sigma}, \quad (2)$$

where the first term represents the single particle Hamiltonian and $V_{\alpha\sigma, \beta\sigma', \mu\sigma', \nu\sigma} = \int d\vec{r} \int d\vec{r}' \varphi_{\alpha\sigma}^*(\vec{r}) \varphi_{\beta\sigma'}^*(\vec{r}') \frac{e^2}{\epsilon|\vec{r} - \vec{r}'|} \varphi_{\mu\sigma'}(\vec{r}') \varphi_{\nu\sigma}(\vec{r})$, is the two-body Coulomb matrix element.

The Hamiltonian for the MIs accounts for MI-MI direct exchange interaction and MI-Zeeman coupling,

$$H_m = \sum_{j,j'=1}^M J_{jj'} \vec{M}_j \cdot \vec{M}_{j'} + \sum_j g_m \mu_b M_{zj} B, \quad (3)$$

where $J_{jj'}$ is the direct MI-MI antiferromagnetic (AFM) coupling, g_m is the MI g-factor, and M_{zj} is the z -component of the MI spin operator.

The nuclear-nuclear direct dipole interaction in the host semiconductor is neglected. The nuclear Hamiltonian is given by the Zeeman coupling term,

$$H_n = \sum_{l=1}^L g_n \mu_b I_{zl} B, \quad (4)$$

where g_n is the nuclear g-factor and I_{zl} is the z -component of nuclear spin operator.

The e-MI exchange interaction is modeled by

$$H_{em} = -J_{em} \sum_{i,j} \vec{S}_i \cdot \vec{M}_j \delta(\mathbf{r}_i - \mathbf{R}_j), \quad (5)$$

with J_{em} the exchange coupling between electron spin $\vec{S}_i$ at $\mathbf{r}_i$ and impurity spin $\vec{M}_j$ located at the position $\mathbf{R}_j$ [10]. In second quantization it can be written as

$$H_{em} = - \sum_{\alpha\beta} \sum_I \frac{J_{\alpha\beta}(\mathbf{R}_j)}{2} [M_{zj} (c_{\alpha\uparrow}^\dagger c_{\beta\uparrow} - c_{\alpha\downarrow}^\dagger c_{\beta\downarrow}) + M_j^+ c_{\alpha\downarrow}^\dagger c_{\beta\uparrow} + M_j^- c_{\alpha\uparrow}^\dagger c_{\beta\downarrow}], \quad (6)$$

where $J_{\alpha\beta}(\mathbf{R}_j) = J_{em} \varphi_{\alpha}^*(\mathbf{R}_j) \varphi_{\beta}(\mathbf{R}_j)$. Similarly, we describe the electron - nuclear spin bath hyperfine interaction by

$$H_{en} = \sum_{i,l} \tilde{A}_l \vec{S}_i \cdot \vec{I}_l \delta(\mathbf{r}_i - \mathbf{R}_l) \quad (7)$$

with $\tilde{A}_l = (16\pi/3)\mu_b\mu_l/I_l$ the isotropic (Fermi contact) part of the electron-nucleus hyperfine interaction [6, 13,